CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in building AI agents with domain experts and taking them from prototype to deployment, with evals, and deterministic code-based extraction where an LLM is not needed. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by consulting work with business stakeholders and AI-assisted development with Claude Code and custom skills.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present | Muldestausee, Germany

- Took end-to-end ownership of the knowledge-extraction pipeline as sole developer, from data ingestion and chunking to prompt engineering and indexing (agentic, deterministic and monolithic strategies), reporting directly to the CTO and working with SAP domain experts across the German energy sector.
- Designed and optimized the multi-agent orchestration in LangGraph (entity-extraction, relationship-analysis, consolidation and reflection agents running in parallel) for complex regulatory and technical documentation, raising precision/recall from 33%/45% to 93%/90% measured against an SAP-system-derived ground truth.
- Engineered a deterministic, code-based extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing accuracy toward 100%.
- Designed and implemented a Model Context Protocol (MCP) service that exposes the knowledge graph for millisecond-latency retrieval, integrated into Claude Code for agentic development workflows.
- Developed Expert Interface, a full-stack graph curation web app (React frontend, Python microservices backend) on Azure where domain experts validate and approve knowledge-graph changes.
- **Tech stack: Python, LangGraph, LangChain, LiteLLM, MCP, Claude Code, Neo4j, GraphRAG, RAG, Vector Search, FastAPI, MongoDB, Postgres, Docker, Kubernetes, Azure, React**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 | Leverkusen, Germany

- Worked with business stakeholders to understand their decision-making workflows, advised on where agentic AI could support them and translated their requirements into the solution approach.
- Designed and deployed an agentic AI application with LangGraph that routes decision-making across vector database queries, general inference and summarization.
- Built the Python/FastAPI backend and a React frontend, tested the APIs with Postman, containerized the stack with Docker and deployed it to AWS with CDK; versioned in GitLab with CI/CD pipelines.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD, Postman, React**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 | Leverkusen, Germany

- Automated evals of RAG generation and retrieval with RAGAS and PytrecEval, against chemist-provided ground truth, with serverless AWS infrastructure for PrismaChat RAG evaluation (Lambda, SQS, DynamoDB) built with CDK.
- Benchmarked multilingual embedding models on over 1M chunks of internal chemical data and designed RAG pipelines (chunking, vector databases, indexing, fine-tuning models for retrieval).
- **Tech stack: Python, RAGAS, PytrecEval, LangChain, Hugging Face Transformers, PyTorch, AWS (Lambda, SQS, DynamoDB, CDK), Azure AI Search, Docker**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM models to predict prepolymer synthesis outcomes from time-series process data, and delivered a Streamlit dashboard for chemists to simulate parameter changes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet and LSTM models for EV load time-series forecasting; the LSTM reached 96% test accuracy with strong performance on unseen data.

PROJECTS

Project: Order Inbox Agent | Python | Django | TypeScript | Postgres | LLMs

- AI agent that reads purchase orders from email bodies, PDFs and Excel files, extracts customer, items and quantities, matches each line to a product catalog (exact, customer map, fuzzy, pgvector) and writes a sales order back to a mock ERP through a REST API.
- Django backend with Postgres and a TypeScript review screen where a sales rep confirms low-confidence matches; per-customer configuration and a labelled test set of orders run in CI.

Project: Agent Eval Harness | Python | LangGraph | pytest | OpenTelemetry | GitHub Actions

- Evaluation harness for a tool-calling LangGraph agent over a mocked API: golden datasets (regression, edge cases, synthetic multi-turn, adversarial), deterministic checks first (JSON schema, function-calling payloads, regex, API status, embedding similarity), then LLM-as-a-judge only for what they cannot decide.
- Runs each case several times and reports pass rates with confidence intervals; tests prompt injection, malformed inputs, unexpected tool responses and tool-loop limits; LLM response caching and tool mocking in GitHub Actions, tracing and observability with OpenTelemetry and Arize Phoenix.

TECHNICAL SKILLSET

- **Agents & Evals:** LangGraph multi-agent workflows, LangChain, LiteLLM, MCP tools, evals (RAGAS, PytrecEval), deterministic code-based extraction, RAG, GraphRAG
- **AI-Assisted Development:** coding agents, Claude Code (custom skills, MCP integration, agentic coding workflows)
- **Customer Delivery:** stakeholder requirements, agentic AI use-case consulting, deployments, domain experts (SAP experts, chemists)
- **Full-Stack & Backend:** Python, FastAPI, Python microservices, SQL, Postman, React (beginner)
- **Data & Knowledge:** Neo4j knowledge graphs, MongoDB, Postgres, vector databases (Qdrant, Pinecone, Azure AI Search), Hugging Face Transformers, PyTorch, Scikit-Learn
- **Cloud & DevOps:** Docker, Kubernetes, Azure, AWS (EC2, Lambda, SQS, DynamoDB, S3, CDK), Git, GitLab CI/CD

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering — August 2017 – July 2021
Minor in Computer Science & Engineering
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)